

Non-functional characteristics and Quality Conflict:

Exam Scenario Question

A multinational company, **TechFusion Systems**, has developed a large-scale **Integrated Emergency Response Software (IERS)** used by hospitals, police departments, and disaster management authorities. The system integrates real-time data processing, GPS tracking, patient record access, communication modules, and predictive analytics.

Recently, users from different departments have reported mixed feedback:

- **Doctors** complain that the system is **slow when retrieving patient data** during emergencies.
- **Field officers** report that the system **crashes occasionally under heavy load**, especially during natural disasters.
- **IT administrators** argue that the system is **highly secure and robust**, as it uses advanced encryption and multiple validation layers.
- **New employees** find the system **extremely difficult to learn and understand**, requiring weeks of training.
- **Developers** claim that the system is **modular and reusable**, but admit that adding new features often introduces unexpected bugs.
- **Management** insists that the system must remain **portable across different devices** and maintain **high security standards** due to sensitive data.

An internal audit reveals:

- Increasing **system complexity** due to continuous feature additions.
- Extensive **security checks** slowing down performance.
- High **coupling between modules**, reducing testability.
- Frequent **trade-offs between reliability and efficiency** during peak usage.
- Lack of a clearly defined **quality assessment process**.

Questions

(a) Identify and explain any **FIVE non-functional quality attributes** from the scenario that are either:

- Achieved, OR
- Problematic

Support each with evidence from the scenario.

(b) Analyze the scenario and identify **TWO major quality conflicts**.

For each conflict:

- Clearly state the conflicting attributes
- Explain *why improving one negatively affects the other* in this system

(c) The organization decides to prioritize **Reliability, Security, and Efficiency** in its quality plan.

Critically evaluate this decision:

- What **trade-offs** will likely occur?
- Which other quality attributes may suffer?

(d) Propose a **quality assessment process** for this system:

- Define how the company can measure whether **Reliability** and **Usability** are achieved
- Suggest **specific metrics or methods**

(e) A junior engineer argues:

“If the system is functional, users will always find a workaround. So non-functional requirements are less important.”

Do you agree or disagree? Justify your answer using the scenario.